

TEST CODE 09630010

FORM TP 2021389

GUYANA
MINISTRY OF EDUCATION
NATIONAL GRADE SIX ASSESSMENT

SCIENCE

Paper 01

AUGUST 4, 2021 (p.m.)

1 hour and 10 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test has 40 questions. You have 1 hour and 10 minutes to answer them.
2. Each question has **four** suggested answers: (A), (B), (C) and (D). Read each question carefully, then choose the correct answer.
3. On your answer sheet, find the number that matches the question you intend to answer.
4. Shade the circle with the letter, (A), (B), (C) or (D), that matches your answer for each question.

Sample Question

Carbon dioxide is a

- (A) mixture.
- (B) liquid.
- (C) solid.
- (D) gas.

Sample Answer



The correct answer is “gas”, so (D) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When the supervisor tells you to begin, turn the page and work as quickly and as carefully as you can.
7. If you try a question and find that you cannot answer it, leave it and go on to the next one. You may return to that question later.
8. You must **not** use calculators for this examination. Rough work may be done in this booklet.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. In which of the following body systems is the kidney an organ?
 - (A) Digestive
 - (B) Excretory
 - (C) Circulatory
 - (D) Respiratory

2. Which two body systems work together to transport oxygen to the cells in the human body?
 - (A) Digestive and skeletal
 - (B) Circulatory and skeletal
 - (C) Respiratory and digestive
 - (D) Circulatory and respiratory

3. Which of the following shows the correct path through which carbon dioxide is eliminated from the human body?
 - (A) Other parts of the body → heart → lungs → nose
 - (B) Other parts of the body → lungs → heart → nose
 - (C) Heart → lungs → other parts of the body → nose
 - (D) Nose → lungs → heart → other parts of the body

4. The main classes of invertebrates are
 - (A) mammals, animals, fishes, birds, reptiles.
 - (B) birds, fishes, reptiles, nematodes, annelids.
 - (C) mammals, fishes, birds, amphibians, reptiles.
 - (D) annelids, molluscs, arthropods, worms, nematodes.

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5. Which of the following animals is **not** an invertebrate?

- (A) Hawk
- (B) Shrimp
- (C) Butterfly
- (D) Octopus

6. The green pigment in leaves that helps them to carry out photosynthesis is called

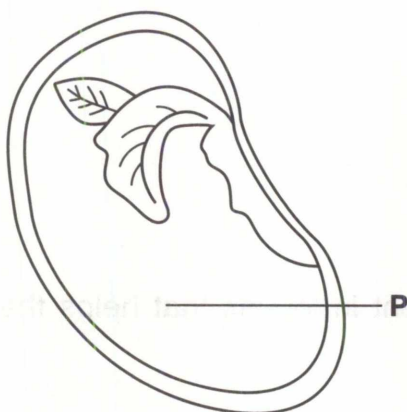
- (A) palisade.
- (B) mesophyll.
- (C) chlorophyll.
- (D) mitochondria.

7. Which part of the plant allows excess water to be given off?

- (A) Fruit
- (B) Root
- (C) Stem
- (D) Leaf

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Question 8 refers to the following diagram which shows a bean seed.



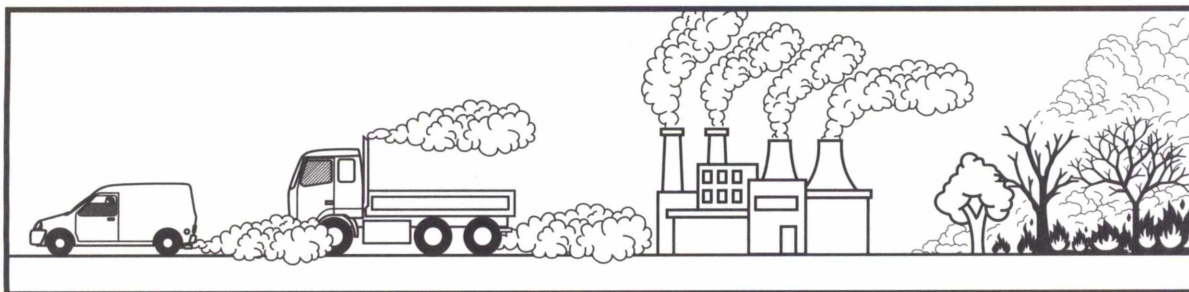
8. The part of the seed labelled P is the

- (A) radicle.
- (B) plumule.
- (C) cotyledon.
- (D) micropyle.

9. Which of the following materials is **non-biodegradable**?

- (A) Paper cup
- (B) Newspaper
- (C) Toilet tissue
- (D) Styrofoam box

Question 10 refers to the following diagram which shows a type of pollution.

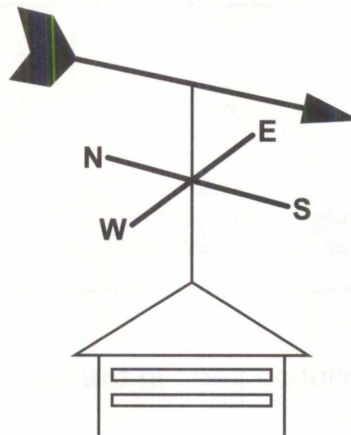


10. The main type of pollution seen in the diagram above is

- (A) air pollution.
- (B) land pollution.
- (C) noise pollution.
- (D) water pollution.

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Question 11 refers to the following diagram which shows a weather instrument.

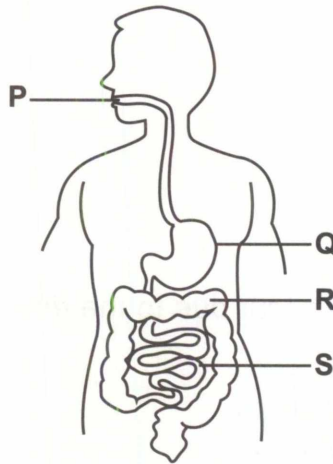


11. The weather instrument shown in the diagram above is
- (A) a barometer.
 - (B) a wind vane.
 - (C) a rain gauge.
 - (D) an anemometer.
12. The breaking down of rocks by the sun, water and wind is called
- (A) flooding.
 - (B) expansion.
 - (C) weathering.
 - (D) contraction.
13. When sugar and milk are added to hot water to make tea, the hot water is referred to as the
- (A) solute.
 - (B) solvent.
 - (C) solution.
 - (D) suspension.

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14. Solids that dissolve in water are said to be
- (A) soluble.
 - (B) miscible.
 - (C) insoluble.
 - (D) immiscible.
15. The phases of the moon occur due to the movement of the
- (A) earth around the sun.
 - (B) moon around the sun.
 - (C) moon around the earth.
 - (D) earth around the moon.
16. The part of the eye that allows light to enter and focus on the retina is called the
- (A) iris.
 - (B) pupil.
 - (C) sclera.
 - (D) cornea.
17. Which of the following statements about the circulatory system is **true**?
- (A) Blood received by the heart always has a low oxygen content.
 - (B) Blood that has a high oxygen content only flows from the lungs to the heart.
 - (C) Blood flowing from other parts of the body to the heart has a higher oxygen content.
 - (D) The exchange of gases only takes place between the blood vessels in the head and legs.

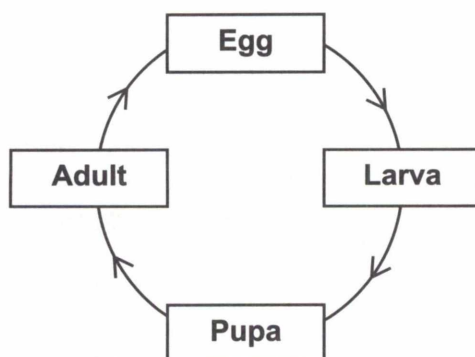
Question 18 refers to the following diagram which shows the human digestive system.



18. Which of the parts labelled P, Q, R and S work closely with the circulatory system to absorb nutrients into the body?
- (A) P
 - (B) Q
 - (C) R
 - (D) S
19. Floyd found a strange animal under a rock. It had soft, moist skin, muscular foot and its body was covered by a mantle and a shell. This animal was **most** likely
- (A) a mollusc.
 - (B) an annelid.
 - (C) a crustacean.
 - (D) a roundworm.

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Question 20 refers to the following diagram which shows the life cycle of an animal.



20. Which group of animals below goes through the life cycle shown above?
- (A) Cockroach, mealworm, spider and bat
 - (B) Butterfly, housefly, beetle and mosquito
 - (C) Housefly, cockroach, frog and mosquito
 - (D) Grasshopper, beetle, dragonfly and frog
21. Which of the following statements about germination and plant growth is correct?
- (A) All seeds need soil and moisture to germinate.
 - (B) Plants grow only from seeds and need soil to germinate.
 - (C) Plants grow from seeds and parts of stems, leaves, and roots.
 - (D) In a seedling, the plumule becomes the root and the radicle becomes the shoot.

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Questions **22** and **23** refer to the following groups of seeds labelled P, Q, R and S.

Group	Seeds
P	Guava, rice, awara
Q	Red beans, peanut, corn
R	Corn, rice, awara
S	Bora, mango, peanut

22. Which group of seeds has one cotyledon?

- (A) P
- (B) Q
- (C) R
- (D) S

23. A plant has net-veined leaves, flowers with five petals and a taproot. The seed of this plant would **most** likely be placed in group

- (A) R
- (B) S
- (C) P
- (D) Q

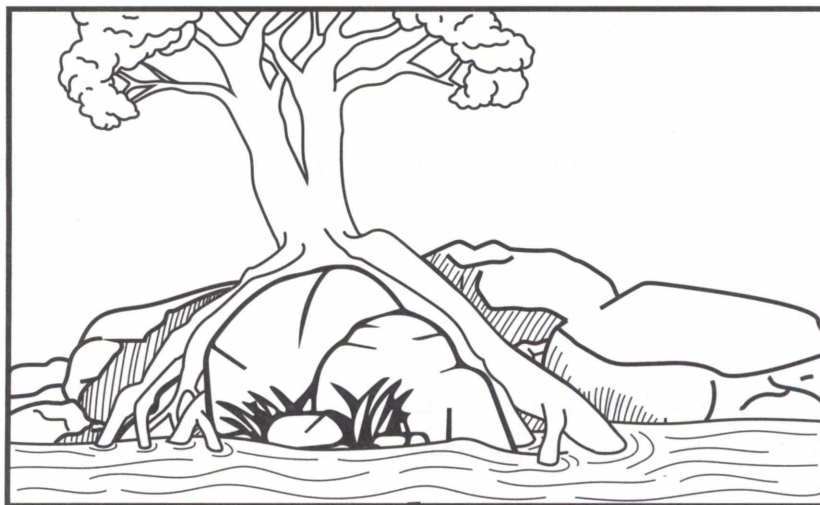
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24. Which of the following situations will cause water pollution?

- I. Dumping of litter in any location
- II. Oil spills and chemicals from factories
- III. Use of excessive agricultural chemicals on the farm

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

Question 25 refers to the following diagram which shows a large rock breaking apart.



25. Which of the following agents of weathering is **most** likely causing the rock to break apart?

- (A) Plant root
- (B) Hot sunlight
- (C) Strong wind
- (D) Running water

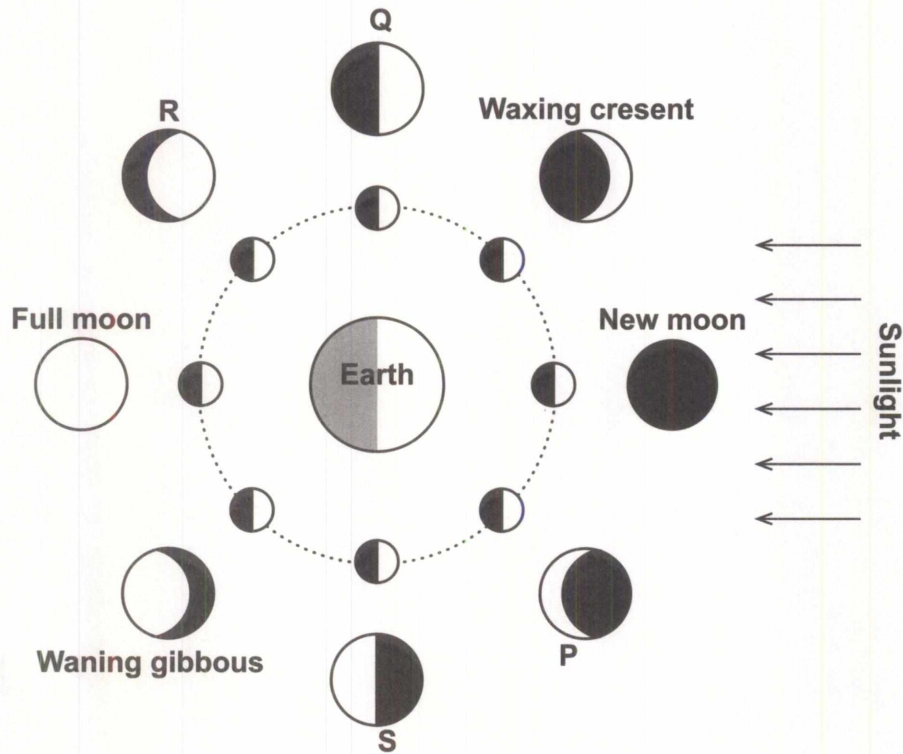
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26. Which of the following techniques would **best** be used to separate soil and water?
- (A) Filtration
 - (B) Decanting
 - (C) Chromatography
 - (D) Fractional distillation
27. In which of the following processes does matter change from a liquid state to a gaseous state?
- (A) Sieving
 - (B) Filtration
 - (C) Evaporation
 - (D) Condensation
28. Which of the following methods would **not** increase the rate at which a solid dissolves?
- (A) Stirring
 - (B) Crushing the solid
 - (C) Increasing the temperature
 - (D) Decreasing the temperature

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NOTHING HAS BEEN OMITTED.

Questions 29 – 30 refer to the following diagram which shows the phases of the moon. The 4 phases labelled P, Q, R and S are missing.



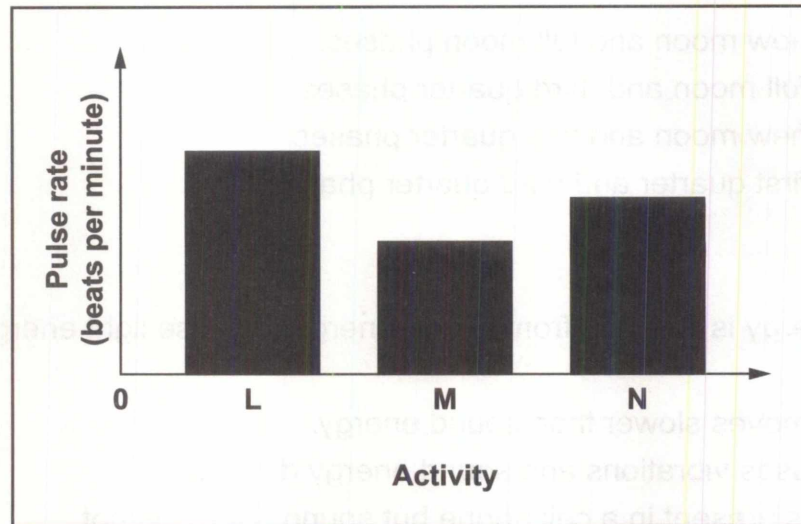
29. Which of the following options represents the correct phases P, Q, R and S?

	P	Q	R	S
(A)	Waxing gibbous	First quarter	Waning crescent	Last quarter
(B)	Waning crescent	First quarter	Waxing gibbous	Last quarter
(C)	Waning crescent	Last quarter	Waxing gibbous	First quarter
(D)	Waxing gibbous	Last quarter	Waning crescent	First quarter

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30. From the earth, the amount of the lighted side of the moon that is seen is the same during the
- (A) new moon and full moon phases.
 - (B) full moon and third quarter phases.
 - (C) new moon and first quarter phases.
 - (D) first quarter and third quarter phases.
31. Light energy is different from sound energy because light energy
- (A) moves slower than sound energy.
 - (B) uses vibrations and sound energy does not.
 - (C) is present in a cell phone but sound energy is not.
 - (D) is visible to the human eye and sound energy moves in waves.
32. Which of the following activities is an example of the use of sound energy?
- (A) Cooking dinner in the oven
 - (B) Putting water in the freezer
 - (C) Listening to someone talking
 - (D) Using a lamp to read a book

Question 33 refers to the following graph which shows Priya's pulse rate immediately after three different activities, **L**, **M** and **N**.

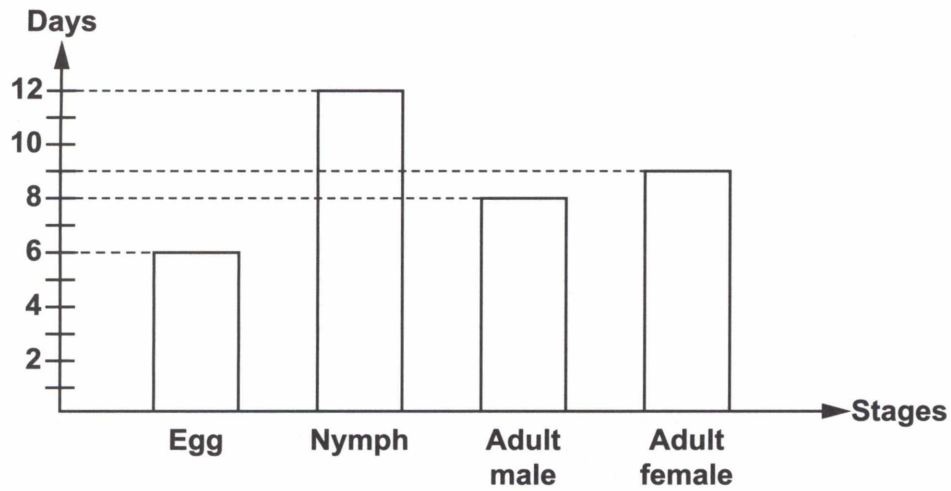


33. Which of the following combinations **most** likely represents the activities, **L**, **M** and **N**?

	L	M	N
(A)	Sprinting	Walking	Climbing
(B)	Jogging	Sleeping	Reading
(C)	Running	Eating	Resting
(D)	Walking	Climbing	Sprinting

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Question 34 refers to the following graph, which shows the number of days in each stage of the life cycle of an insect.

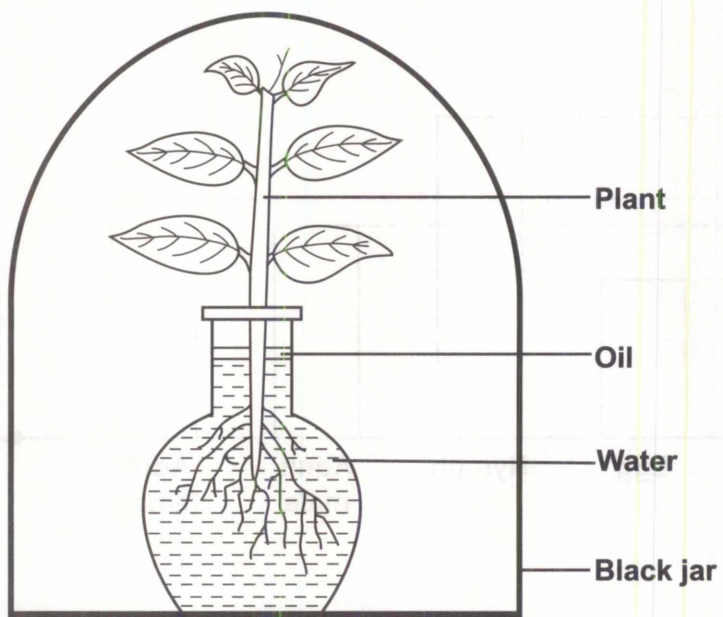


34. **Altogether**, for this particular insect, the number of days in the life cycle of an adult male is

- (A) 8 days
- (B) 12 days
- (C) 18 days
- (D) 26 days

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Question 35 refers to the following illustration of an experiment. The set-up was left in the sun for a few hours.



35. Which of the following processes **will** likely occur in the container?

- I. Germination
- II. Transpiration
- III. Photosynthesis

- (A) II only
- (B) III only
- (C) I and II only
- (D) I, II and III

36. To prevent pollution, proper and safe methods of waste disposal must be implemented. Which of the following methods is **not** a safe way to dispose of waste?

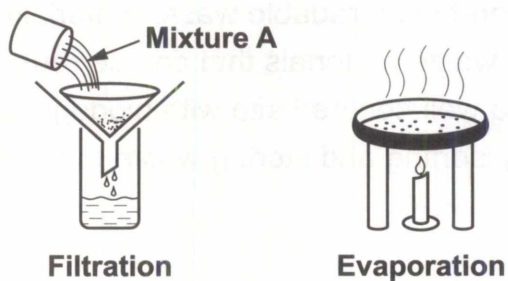
- (A) Burning non-biodegradable waste materials
- (B) Recycling waste materials that can be reused
- (C) Filling up a well-secured site with biodegradable waste
- (D) Collecting, sorting and storing waste materials properly

37. Which of the following statements about weathering is **false**?

- (A) Rocks of different compositions weather at different rates.
- (B) Heat and heavy rainfall increase the rate of physical weathering.
- (C) The presence of soil increases weathering of the rocks underneath it.
- (D) The longer a rock is exposed at the surface, the more weathering occurs.

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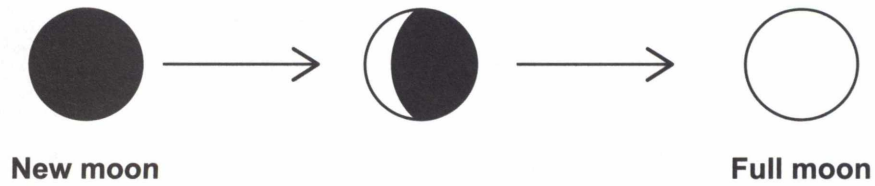
Question 38 refers to the following diagrams, which show the two techniques used to separate two substances, **X** and **Y**, from Mixture **A**. Mixture **A** was first filtered to obtain **X**. The filtrate was then evaporated to obtain **Y**.



38. Which two substances were **most** likely separated from Mixture **A**?

	X	Y
(A)	Sugar	Salt
(B)	Water	Salt
(C)	Sand	Salt
(D)	Sand	Water

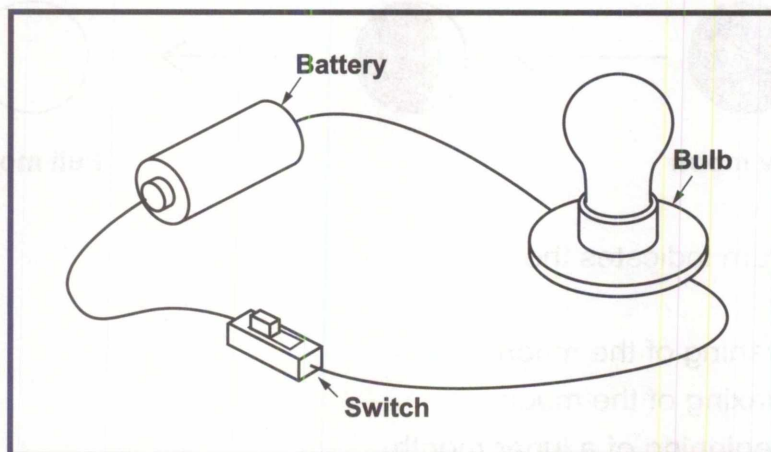
Question 39 refers to the following diagram of the phases of the moon.



39. The diagram indicates the
- (A) waning of the moon.
 - (B) waxing of the moon.
 - (C) beginning of a lunar month.
 - (D) formation of a lunar eclipse.

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Question 40 refers to the following diagram of a simple circuit.



40. What is the **correct** energy conversion that occurs in lighting the bulb, as illustrated in the diagram above?
- (A) Light → electrical → chemical
 - (B) Chemical → electrical → light
 - (C) Electrical → light → chemical
 - (D) Light → chemical → electrical

END OF TEST

IF YOU FINISH BEFORE TIME IS UP, CHECK YOUR WORK ON THIS TEST.

**DO NOT
WRITE ON
THIS PAGE**

THE CARIBBEAN EXAMINATIONS COUNCIL

Prepared by

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